

North South University

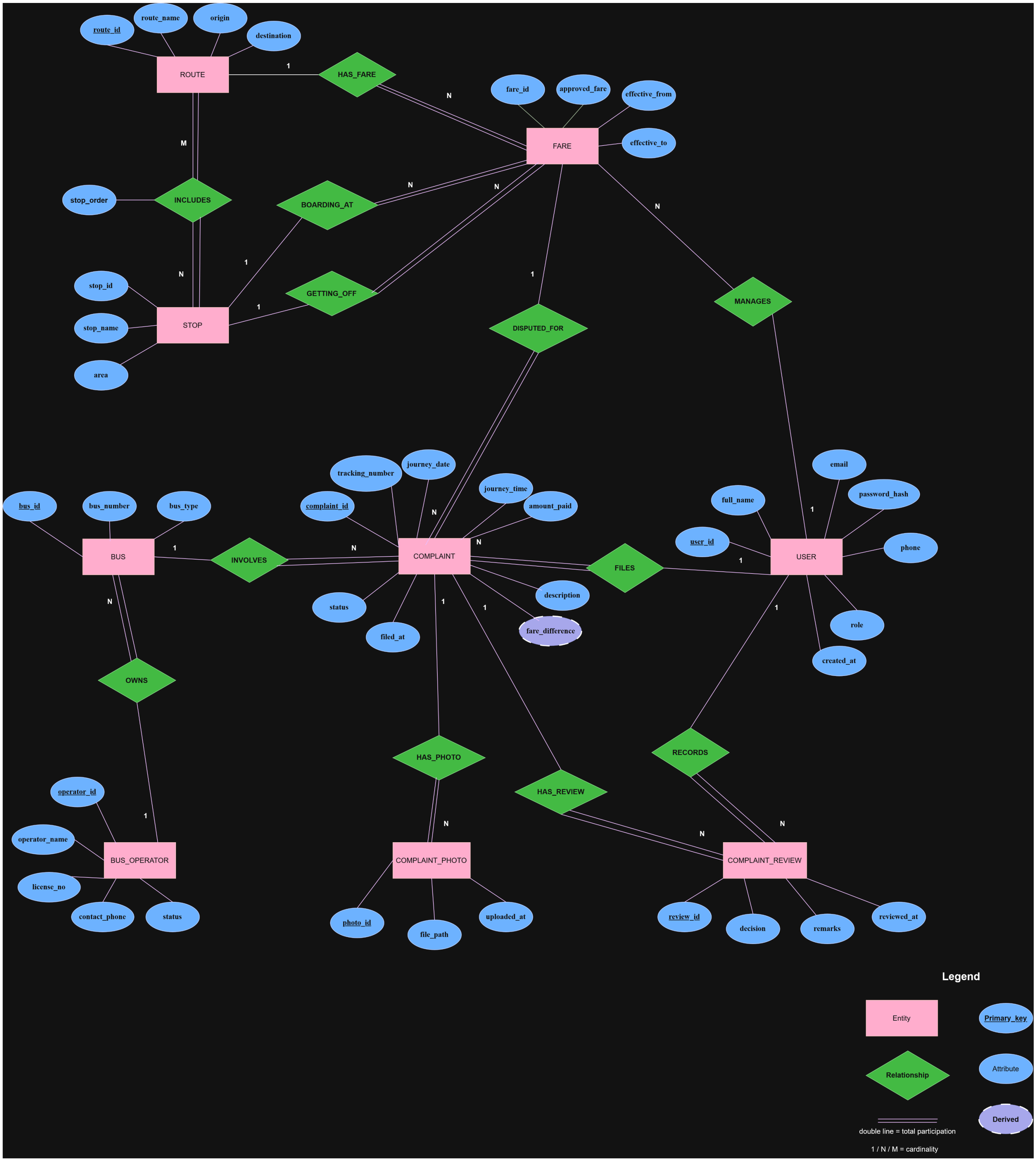
Department of Electrical & Computer Engineering

ERD & SCHEMA DESIGN

Course Code	CSE311L
Course Title	Database Systems Lab
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Section	3
Group / Team Number	Team 6
Project Title	Public Transport Fare Dispute Redressal System
Submission	ERD & Schema Design
Date of Submission	05/11/26
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ER Diagram (Chen Notation)

Public Transport Fare Dispute Redressal System



1. List of Entities & Attributes

The design has **9 entities** plus one junction table (ROUTE_STOP) created from the M:N relationship INCLUDES, so the logical schema has **10 tables**. Primary keys are marked **PK** and foreign keys **FK**. In the Chen ER diagram foreign keys are shown as relationships, so they appear here and in Section 3 instead of as ovals.

USER

Passengers and admins who log in to the system (role decides the available features).

Attribute	Data Type / Constraint	Key	Description
user_id	INT, AUTO_INCREMENT	PK	Unique ID of the user
full_name	VARCHAR(100), NOT NULL		Full name
email	VARCHAR(100), NOT NULL, UNIQUE		Login email
password_hash	VARCHAR(255), NOT NULL		Hashed password (never plain text)
phone	VARCHAR(15)		Contact number
role	ENUM('Passenger','Admin'), NOT NULL, DEFAULT 'Passenger'		Decides access rights
created_at	TIMESTAMP, DEFAULT NOW		Registration time

BUS_OPERATOR

Company or owner that runs buses; admins track repeated complaints against it.

Attribute	Data Type / Constraint	Key	Description
operator_id	INT, AUTO_INCREMENT	PK	Unique ID of the operator
operator_name	VARCHAR(100), NOT NULL		Operator / company name
license_no	VARCHAR(50), NOT NULL, UNIQUE		Operating license number
contact_phone	VARCHAR(15)		Operator contact number
status	ENUM('Active','Suspended'), NOT NULL, DEFAULT 'Active'		Current operating status

BUS

An individual bus owned by one operator.

Attribute	Data Type / Constraint	Key	Description
bus_id	INT, AUTO_INCREMENT	PK	Unique ID of the bus
bus_number	VARCHAR(20), NOT NULL, UNIQUE		Registration / bus number
bus_type	VARCHAR(30)		e.g. Local, AC, Express
operator_id	INT, NOT NULL	FK	Owning operator → BUS_OPERATOR

ROUTE

A bus route with its approved start and end points.

Attribute	Data Type / Constraint	Key	Description
route_id	INT, AUTO_INCREMENT	PK	Unique ID of the route
route_name	VARCHAR(100), NOT NULL, UNIQUE		Route name / number
origin	VARCHAR(100), NOT NULL		Starting point
destination	VARCHAR(100), NOT NULL		Ending point

STOP

A bus stop that can appear on many routes.

Attribute	Data Type / Constraint	Key	Description
stop_id	INT, AUTO_INCREMENT	PK	Unique ID of the stop
stop_name	VARCHAR(100), NOT NULL		Name of the stop
area	VARCHAR(100)		Area / locality of the stop

ROUTE_STOP

Junction table for the M:N relationship INCLUDES between ROUTE and STOP.

Attribute	Data Type / Constraint	Key	Description
route_id	INT	PK, FK	Route → ROUTE
stop_id	INT	PK, FK	Stop → STOP
stop_order	INT, NOT NULL		Position of the stop along the route (UNIQUE per route)

FARE

Approved fare chart entry: fare between a boarding and a destination stop on a route.

Attribute	Data Type / Constraint	Key	Description
fare_id	INT, AUTO_INCREMENT	PK	Unique ID of the fare record
route_id	INT, NOT NULL	FK	Route → ROUTE
from_stop_id	INT, NOT NULL	FK	Boarding stop → STOP
to_stop_id	INT, NOT NULL	FK	Destination stop → STOP
approved_fare	DECIMAL(8,2), NOT NULL		Official fare in BDT (CHECK ≥ 0)
effective_from	DATE, NOT NULL		Date this fare starts
effective_to	DATE		Date this fare ends (NULL = current)
managed_by	INT, NOT NULL	FK	Admin who recorded it → USER

COMPLAINT

An overcharging report filed by a passenger.

Attribute	Data Type / Constraint	Key	Description
complaint_id	INT, AUTO_INCREMENT	PK	Unique ID of the complaint
tracking_number	VARCHAR(20), NOT NULL, UNIQUE		Public tracking number
passenger_id	INT, NOT NULL	FK	Reporting passenger → USER
bus_id	INT, NOT NULL	FK	Bus involved → BUS
fare_id	INT, NOT NULL	FK	Fare record for route and stops → FARE
journey_date	DATE, NOT NULL		Date of journey
journey_time	TIME		Time of journey
amount_paid	DECIMAL(8,2), NOT NULL (CHECK > 0)		Amount the passenger actually paid
description	TEXT		Passenger statement
status	ENUM('Pending','Under Review','Resolved','Rejected'), NOT NULL, DEFAULT 'Pending'		Current status (default Pending)
filed_at	TIMESTAMP, DEFAULT NOW		Filing time
fare_difference	amount_paid − FARE.approved_fare (computed in queries, not stored)	derived	Difference between amount paid and approved fare

COMPLAINT_PHOTO

Optional photo evidence attached to a complaint.

Attribute	Data Type / Constraint	Key	Description
photo_id	INT, AUTO_INCREMENT	PK	Unique ID of the photo
complaint_id	INT, NOT NULL	FK	Complaint → COMPLAINT
file_path	VARCHAR(255), NOT NULL		Stored image path
uploaded_at	TIMESTAMP, DEFAULT NOW		Upload time

COMPLAINT_REVIEW

One admin action or decision on a complaint (keeps the full case history).

Attribute	Data Type / Constraint	Key	Description
review_id	INT, AUTO_INCREMENT	PK	Unique ID of the review entry
complaint_id	INT, NOT NULL	FK	Complaint → COMPLAINT
admin_id	INT, NOT NULL	FK	Reviewing admin → USER
decision	ENUM('Under Review','Resolved','Rejected'), NOT NULL		Decision recorded
remarks	TEXT		Admin notes / reason
reviewed_at	TIMESTAMP, DEFAULT NOW		Time of decision

2. Relationships & Cardinality

Cardinality is read as (left entity) : (right entity). Total participation is shown with a double line in the ERD.

Relationship	Entity A	Entity B	Cardinality	Participation	Meaning
FILES	USER	COMPLAINT	1 : N	COMPLAINT total	One passenger can file many complaints; each complaint is filed by exactly one passenger.
MANAGES	USER (admin)	FARE	1 : N	FARE total (managed_by)	An admin records many fare records; each fare record is recorded by one admin.
RECORDS	USER (admin)	COMPLAINT_REVIEW	1 : N	REVIEW total	An admin records many decisions; each decision is made by one admin.
INCLUDES	ROUTE	STOP	M : N	Both total	A route has many stops and a stop can be on many routes. Attribute: stop_order. Resolved into ROUTE_STOP.
HAS_FARE	ROUTE	FARE	1 : N	FARE total	A route has many fare records; each fare record belongs to one route.
BOARDING_AT	STOP	FARE	1 : N	FARE total; STOP partial	A stop can be the boarding stop of many fare records; each fare record has one boarding stop.
GETTING_OFF	STOP	FARE	1 : N	FARE total; STOP partial	A stop can be the destination stop of many fare records; each fare record has one destination stop.
DISPUTED_FOR	FARE	COMPLAINT	1 : N	COMPLAINT total	A fare record can be disputed in many complaints; a complaint refers to one fare record (route + stops).
INVOLVES	BUS	COMPLAINT	1 : N	COMPLAINT total	A bus can be reported in many complaints; each complaint names one bus.
OWNS	BUS_OPERATOR	BUS	1 : N	BUS total	An operator owns many buses; each bus belongs to one operator.
HAS_PHOTO	COMPLAINT	COMPLAINT_PHOTO	1 : N	PHOTO total	A complaint can have many (optional) photos; each photo belongs to one complaint.
HAS_REVIEW	COMPLAINT	COMPLAINT_REVIEW	1 : N	REVIEW total	A complaint can have many review entries; each entry belongs to one complaint.

Summary: 11 one-to-many (1:N) relationships and 1 many-to-many (M:N) relationship (ROUTE–STOP, resolved into ROUTE_STOP). There are no one-to-one relationships.

3. Foreign Keys

#	Child Table.Column	Parent Table.Column	ON DELETE	Relationship
1	BUS.operator_id	BUS_OPERATOR.operator_id	RESTRICT	OWNS
2	ROUTE_STOP.route_id	ROUTE.route_id	CASCADE	INCLUDES
3	ROUTE_STOP.stop_id	STOP.stop_id	RESTRICT	INCLUDES
4	FARE.route_id	ROUTE.route_id	RESTRICT	HAS_FARE
5	FARE.from_stop_id	STOP.stop_id	RESTRICT	BOARDING_AT
6	FARE.to_stop_id	STOP.stop_id	RESTRICT	GETTING_OFF
7	FARE.managed_by	USER.user_id	RESTRICT	MANAGES
8	COMPLAINT.passenger_id	USER.user_id	RESTRICT	FILES
9	COMPLAINT.bus_id	BUS.bus_id	RESTRICT	INVOLVES
10	COMPLAINT.fare_id	FARE.fare_id	RESTRICT	DISPUTED_FOR
11	COMPLAINT_PHOTO.complaint_id	COMPLAINT.complaint_id	CASCADE	HAS_PHOTO
12	COMPLAINT_REVIEW.complaint_id	COMPLAINT.complaint_id	CASCADE	HAS_REVIEW
13	COMPLAINT_REVIEW.admin_id	USER.user_id	RESTRICT	RECORDS

All foreign keys use **ON UPDATE CASCADE**. CASCADE is used only for dependent rows (route stops, photos, review history); everything else is RESTRICT so complaint history and fare records cannot be lost by accident.

4. Logical Schema

Format: TableName(attr1 [PK], attr2, attr3 [FK → OtherTable])

USER(user_id [**PK**], full_name, email, password_hash, phone, role, created_at)

BUS_OPERATOR(operator_id [**PK**], operator_name, license_no, contact_phone, status)

BUS(bus_id [**PK**], bus_number, bus_type, operator_id [**FK → BUS_OPERATOR.operator_id**])

ROUTE(route_id [**PK**], route_name, origin, destination)

STOP(stop_id [**PK**], stop_name, area)

ROUTE_STOP(route_id [**PK, FK → ROUTE.route_id**], stop_id [**PK, FK → STOP.stop_id**], stop_order)

FARE(fare_id [**PK**], route_id [**FK → ROUTE.route_id**], from_stop_id [**FK → STOP.stop_id**], to_stop_id [**FK → STOP.stop_id**], approved_fare, effective_from, effective_to, managed_by [**FK → USER.user_id**])

COMPLAINT(complaint_id [**PK**], tracking_number, passenger_id [**FK → USER.user_id**], bus_id [**FK → BUS.bus_id**], fare_id [**FK → FARE.fare_id**], journey_date, journey_time, amount_paid, description, status, filed_at)

COMPLAINT_PHOTO(photo_id [**PK**], complaint_id [**FK → COMPLAINT.complaint_id**], file_path, uploaded_at)

COMPLAINT_REVIEW(review_id [**PK**], complaint_id [**FK → COMPLAINT.complaint_id**], admin_id [**FK → USER.user_id**], decision, remarks, reviewed_at)

Additional constraints

- **USER.role** ∈ {Passenger, Admin}; **FARE.managed_by** and **COMPLAINT_REVIEW.admin_id** must reference a user whose role is Admin (enforced by a trigger).
- **FARE:** CHECK (approved_fare ≥ 0), CHECK (from_stop_id ≠ to_stop_id), UNIQUE (route_id, from_stop_id, to_stop_id, effective_from). Both stops must belong to the route (checked through ROUTE_STOP). The model does not link BUS to ROUTE, so the bus named in a complaint is recorded as reported by the passenger; the route comes from the FARE record.
- **ROUTE_STOP:** composite primary key (route_id, stop_id) and UNIQUE (route_id, stop_order).
- **COMPLAINT:** tracking_number is generated automatically (trigger) and is UNIQUE; amount_paid > 0; status defaults to Pending. When an admin inserts a COMPLAINT_REVIEW row, a trigger updates COMPLAINT.status. A trigger also checks that journey_date falls between the linked FARE record's effective_from and effective_to (NULL = still current), so a complaint cannot point to a fare that was not in force on the journey date. COMPLAINT.status is kept on the complaint on purpose, for quick tracking lookups; COMPLAINT_REVIEW keeps the full decision history.
- **fare_difference** is a derived attribute (amount_paid − approved_fare). It is calculated in queries and views, so it is not stored and cannot become inconsistent.

5. Consistency with the Project Proposal

Proposal feature	Supported by
1. User Accounts	USER (role = Passenger / Admin)
2. Fare Lookup	ROUTE, STOP, ROUTE_STOP, FARE (route + boarding stop + destination stop → approved_fare)
3. Dispute Filing	COMPLAINT (route/stops via FARE, bus via BUS, amount_paid, journey date/time), COMPLAINT_PHOTO (optional photos)
4. Complaint Tracking	COMPLAINT.tracking_number (UNIQUE) and COMPLAINT.status: Pending / Under Review / Resolved / Rejected
5. Admin Dashboard	Admins manage ROUTE, STOP, BUS_OPERATOR, BUS, FARE; compare COMPLAINT.amount_paid with FARE.approved_fare; record decisions in COMPLAINT_REVIEW
6. Reports and Analytics	Totals and fare differences from COMPLAINT + FARE; disputed routes via FARE.route_id; repeat offenders via COMPLAINT → BUS → BUS_OPERATOR